

Virtual Enterprise Network Defence Laboratory

pfSense, Snort, Segmentation, Captive Portal and Traffic Shaping

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Portfolio Case Study - Detailed Project Oversight and Completion Record

All offensive-security activities described in this document were performed in authorized, isolated academic laboratory environments.

1. Project Overview

A defence-in-depth virtual enterprise network project that deployed pfSense as a perimeter firewall, Snort as a network intrusion detection system, source-based segmentation, captive-portal authentication, HFSC traffic shaping, and a deliberately vulnerable target used to evaluate residual host risk.

Project Field	Details
Role and Responsibility	Network designer, firewall/IDS administrator, security tester, analyst, and report author.
Environment and Tools	Oracle VirtualBox, pfSense 2.7.2, Snort, Kali Linux, Windows 10, Typhoon vulnerable VM, isolated LAN/WAN segments.
Completion Status	Completed - documented, tested, and included in the master portfolio.
Portfolio Classification	Academic / personal technical project with recruiter-facing case-study documentation.

2. Project Objectives and Success Criteria

- Design a routed and segmented virtual network with a protected LAN.
- Configure stateful firewall policy and validate connectivity.
- Deploy Snort rules and verify alert generation.
- Demonstrate source-based isolation and captive-portal enforcement.
- Configure HFSC traffic shaping and a penalty queue.
- Evaluate residual risk through authorized reconnaissance and exploitation.

3. Project Oversight and Execution Process

3.1 Planning and Scope

Defined the authorized environment, project goal, tools, expected outputs, and safety limitations.

3.2 Implementation

Executed the technical tasks and maintained notes, commands, screenshots, and configuration evidence.

3.3 Validation and Testing

Compared actual output with objectives, confirmed repeatable behavior, and collected proof of completion.

3.4 Analysis and Reporting

Converted technical evidence into findings, defensive interpretation, limitations, and recommendations.

3.5 Portfolio Finalization

Prepared a recruiter-friendly summary, detailed case study, screenshots, source report, and public-release guidance.

4. Technical Methodology

1. Assigned WAN and LAN interfaces and created a documented addressing scheme.
2. Validated baseline connectivity from Windows, Kali, and the vulnerable target.
3. Reviewed and modified LAN/WAN firewall rules using least-privilege principles.
4. Performed Nmap reconnaissance to inventory exposed services.
5. Installed Snort with community and Emerging Threats rules and reviewed alerts.
6. Added source-based blocking, captive portal authentication, and traffic shaping.
7. Compared perimeter controls with host-level weaknesses and produced hardening guidance.

5. Deliverables and Completion Evidence

Deliverable	Status	Evidence
Technical implementation / lab execution	Complete	Commands, configurations, code, or forensic workflow documented.
Testing and validation	Complete	Results and screenshots included below and in the original report.
Detailed project documentation	Complete	This case-study document and README.
Source/original evidence	Complete	Original report or source files stored in this project folder.
Recruiter-ready description	Complete	Summary and skills section

		included.
Public publishing review	Required before posting	Remove credentials, personal identifiers, confidential case data, and unnecessary target details.

6. Results and Outcomes

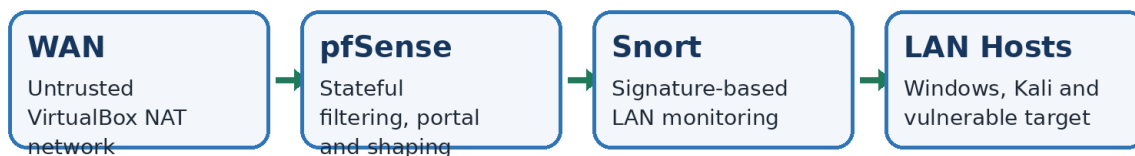
- The network successfully enforced stateful filtering, authentication, source isolation, intrusion visibility, and bandwidth governance.
- Snort generated alerts from observed traffic and firewall rules blocked the designated source as intended.
- The assessment demonstrated that perimeter controls cannot compensate for a host exposing numerous outdated or weakly configured services.

7. Challenges, Decisions, and Lessons Learned

- Maintaining administrative access while tightening firewall rules.
- Separating detection evidence from benign traffic and false positives.
- Explaining the interaction between perimeter security and vulnerable host services.

8. Screenshots and Supporting Evidence

Layered Virtual Network Security Architecture



Architecture diagram of the virtualized enterprise lab.

Recommended Portfolio Presentation

Public Website

Display the three project summaries as individual case-study cards. Each card should include a concise overview, technology tags, one architecture image, two or three evidence screenshots, a “What I learned” section, and a link to an approved supporting artifact. Publish the DDoS documentation and cleaned source code. Publish the network-security report only after removing any unnecessary personal or lab-sensitive information. Do not publish the confidential forensic report without permission.

GitHub Repository Structure

```
ddos-detection-mitigation/
├── README.md
├── src/ddos_detector.py
├── docs/project-case-study.pdf
├── media/demo-video-link.txt
├── screenshots/
└── LICENSE
```

Resume Project Section

DDoS Detection and Mitigation System

Python, Scapy, Matplotlib, Linux, iptables — Built a real-time SYN-flood detector with baseline learning, visualization, alerting, logging, and automated source blocking; verified using an authorized virtual-lab demonstration.

Virtual Enterprise Network Defence Lab

pfSense, Snort, Kali Linux, VirtualBox — Implemented layered firewalling, IDS monitoring, authentication and traffic shaping, then evaluated residual risk through controlled reconnaissance and exploitation.

Academic Digital Forensics Investigation

Autopsy, FTK Imager — Verified evidence integrity and analyzed deleted files, metadata, email and timelines across multiple forensic images; produced a structured confidential report.

Next Projects to Add

Future lab assignments can be added using the same template: overview, problem, objectives, environment, implementation, evidence, results, challenges, improvements, skills, and a resume-ready bullet. The strongest additions will be projects that demonstrate independent design decisions, measurable outcomes, and clean documentation rather than routine screenshots alone.

[Portfolio evidence page illustrating network implementation and security controls.](#)

control: first validate the network, then introduce restrictions, and finally verify the effect of each rule.

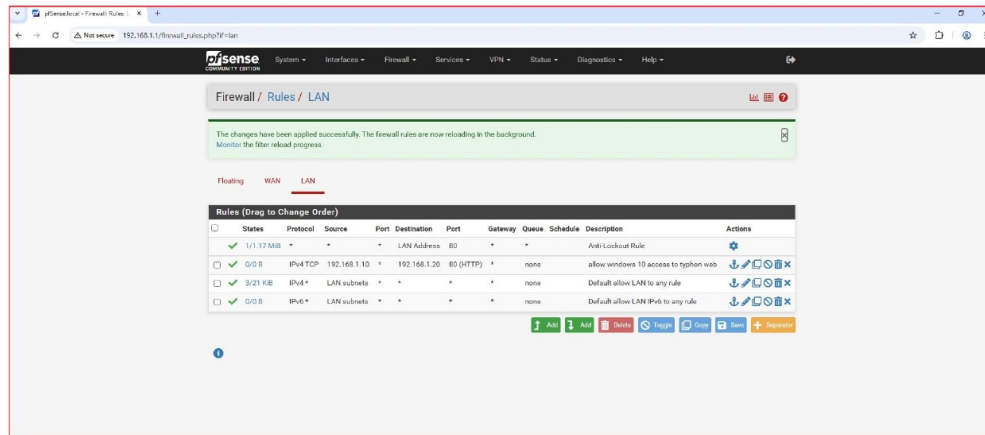


Figure 7. Baseline pfSense LAN policy showing the anti-lockout rule, a specific HTTP permit, and the initial LAN-to-any rules.

Phase 2 - Reconnaissance and Attack-Surface Analysis

Nmap service and operating-system detection identified a broad set of exposed services, including FTP, SSH, SMTP, DNS, HTTP, RPC/NFS, Samba, databases, printing services, and Tomcat. Version disclosure and insecure defaults created a clear picture of the target attack surface. The assessment did not stop at listing open ports; each service was interpreted in terms of likely business impact, authentication exposure, and hardening priority.

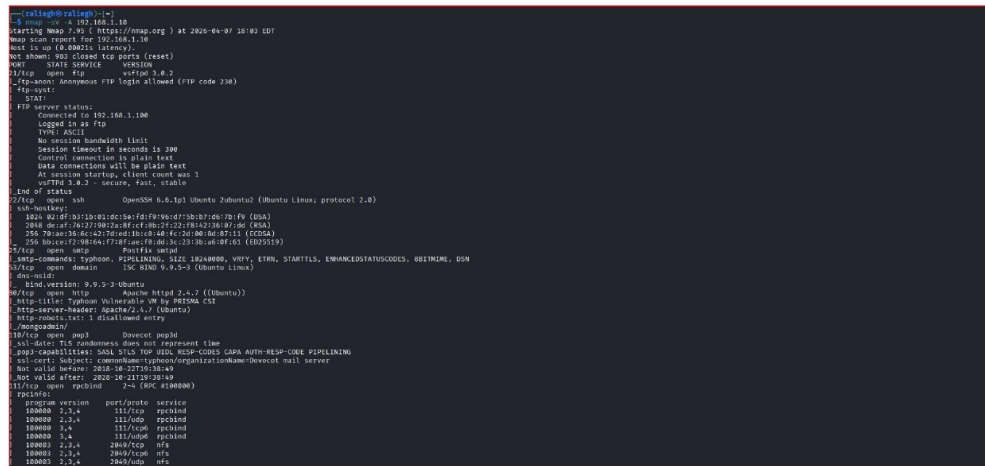


Figure 8. First portion of the Nmap service scan, providing evidence of multiple exposed services and software versions on the vulnerable target.

Portfolio evidence page showing later implementation/testing results.

9. Skills Demonstrated

pfSense, Snort, firewall policy, network segmentation, IDS analysis, Nmap, captive portals, HFSC traffic shaping, VirtualBox, defence-in-depth reporting.

10. Recommendations and Future Improvements

- Create dedicated VLANs for users, servers, management, and security monitoring.
- Use inline IPS where appropriate, centralized logging, and regular rule tuning.
- Patch or retire vulnerable services and enforce endpoint hardening.
- Add identity-aware access and automated configuration backups.

11. Portfolio and Resume Presentation

Virtual Enterprise Network Defence Laboratory - A defence-in-depth virtual enterprise network project that deployed pfSense as a perimeter firewall, Snort as a network intrusion detection system, source-based segmentation, captive-portal authentication, HFSC traffic shaping, and a deliberately vulnerable target used to evaluate residual host risk.

Recommended repository contents: README.md, Documentation/, Evidence/, Source/ or Original_Report/, and a clear authorization/privacy notice.

12. Completion Statement

This project is considered complete for portfolio use because the scope, implementation process, testing evidence, results, skills, limitations, and future improvements are documented. Before public release, sensitive data must be reviewed and removed.